**Suggested teaching order for**

# IB Mathematics AA SL

**Teacher: Mr. Leo**

**Topics**

| | |
|---|---|
| Topic 4: Statistics and Probability | 4.1—4.11 |
| Topic 1: Number and Algebra | 1.1—1.7 |
| Topic 2: Functions | 2.1—2.10 |
| Topic 3: Geometry and Trigonometry | 3.1—3.7 |
| Topic 5: Calculus | 5.1—5.11 |

(paragraphs are according to the syllabus components)

Plan

| Year 1 | Year 2 |
|---|---|
| 70% of the syllabus | 30% of the syllabus |

| YEAR 1 | | |
|---|---|---|
| **Topic 4** | **Statistics and Probability** | **~27 Teaching hours** |
| **4.1** | Basic concepts of Statistics | |
| **4.2** | Measures of central tendency – Measures of spread | |
| **4.3** | Frequency tables – Grouped data | |
| **4.4** | Regression | |
| **4.5** | Elementary set theory | |
| **4.6** | Probability | |
| **4.7** | Conditional probability – independent events | |
| **4.8** | Tree diagrams | |
| **4.9** | Distributions – Discrete random variables | |
| **4.10** | Binomial distribution – $B(n, p)$ | |
| **4.11** | Normal distribution – $N(\mu,\ \sigma)$ | |
| | **Midterm (~October/November)** | |

| | | |
|---|---|---|
| **Topic 1** | **Number and Algebra** | **~19 Teaching hours** |
| **1.1** | Numbers – rounding – scientific form | |

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| | | |
|---|---|---|
| **1.2** | Sequences in general – Series | |
| **1.3** | Arithmetic sequences | |
| **1.4** | Geometric sequences | |
| **1.5** | Applications of G.S. – Percentage growth | |
| **1.6** | The binomial theorem | |
| **1.7** | Deductive proof | |

| **Topic 2** | **Functions** | **$\sim 21$ Teaching hours** |
|---|---|---|
| **2.1** | Lines (or linear functions) | |
| **2.2** | Quadratics (or quadratic functions) | |
| **2.3** | Functions: domain, range and graph | |
| **2.4** | Composite functions: $f \circ g$ | |
| **2.5** | Inverse functions: $f^{-1}$ | |
| | Revision 1 | |
| | **End of term (~January)** | |
| **2.6** | Transformations of functions | |
| **2.7** | Asymptotes | |
| **2.8** | The exponential function | |
| **2.9** | The logarithmic function | |
| **2.10** | Exponential equations | |

| **Topic 3** | **Geometry and Trigonometry** | **$\sim 25$ Teaching hours** |
|---|---|---|
| **3.1** | 3D geometry | |
| **3.2** | Triangles – sine and cosine rules | |
| **3.3** | Applications in 3d geometry – navigation | |
| **3.4** | The trigonometric circle – arcs and sectors | |
| **3.5** | $\sin\theta$, $\cos\theta$, $\tan\theta$ on the unit circle | |
| | **Midterm (~April)** | |
| **3.6** | Trigonometric identities and equations | |
| **3.7** | Trigonometric functions | |

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| YEAR 2 | | |
|---|---|---|
| **Topic 5** | **Calculus** | **$\sim 28$ Teaching hours** |
| **5.1** | The limit $\lim f(x)$ – the derivative $f'(x)$: a rough idea | |
| **5.2** | Derivatives of known functions – rules | |
| **5.3** | Tangent line and normal line at some point $x_0$ | |
| **5.4** | The chain rule | |
| **5.5** | Monotony – max and min | |
| | **Midterm ($\sim$October/November)** | |
| **5.6** | Concavity – points of inflection | |
| **5.7** | Optimisation | |
| **5.8** | The indefinite integral | |
| **5.9** | Integration by substitution | |
| **5.10** | The definite integral – areas between curves | |
| **5.11** | Kinematics (displacement, velocity, acceleration) | |
| | Revision 2 | |
| | **Mock ($\sim$April)** | |

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